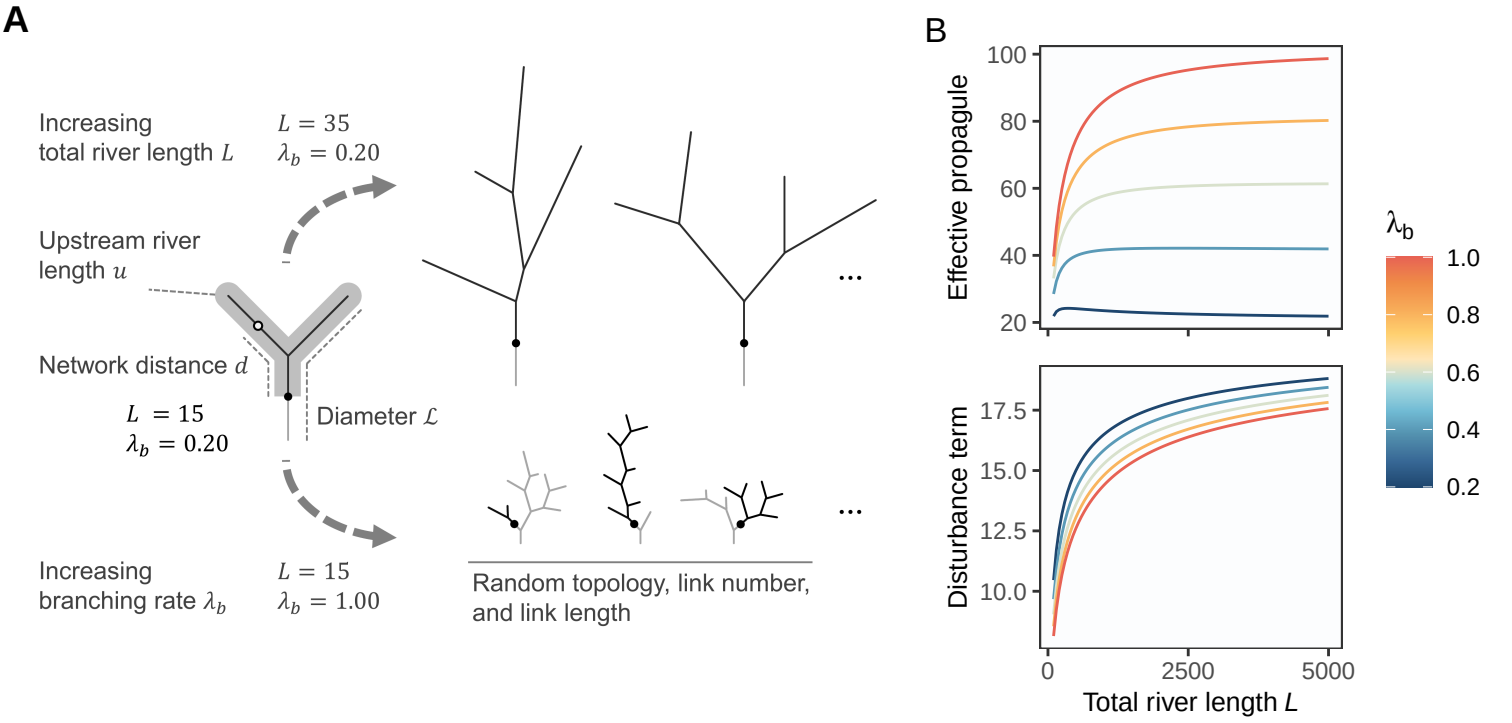

A
Increasing
total river length L
$L = 35$
$\lambda_b = 0.20$
Upstream river
length u
Network distance d
$L = 15$
$\lambda_b = 0.20$
Diameter $\mathcal{L}$
Increasing
branching rate λ_b
$L = 15$
$\lambda_b = 1.00$
Random topology, link number,
and link length
B
Effective propagule
Disturbance term
Total river length L
λ_b
1.0
0.8
0.6
0.4
0.2